

Career Opportunities

اس باب کی مہارتیں پائتھن ڈویلپر، ڈیٹا اینالسٹ، آٹومیشن انجینئر اور گیم ڈویلپر کے کام میں روز استعمال ہوتی ہیں۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 18 of 18

Career Opportunities

LEARN IT

The skills in this chapter — data structures, functions and files — are the same skills used in many jobs.

- **Python / software developer** — builds applications, scripts and tools in Python, one of the most widely used languages in industry.
- **Data analyst** — uses lists and dictionaries, then file handling, to process, visualise and interpret data for decisions.
- **Automation engineer** — writes scripts that repeat file and testing tasks so humans do not have to.
- **Game developer** — uses functions, data structures and logic to build interactive games.

WORKED EXAMPLE

The word-frequency dictionary is a tiny version of what a data analyst does with a much larger file of customer comments. The grade tracker is a tiny version of a school information system. The attendance list is a tiny version of an HR register. Same structures, larger files.

REMEMBER IT

Four careers: **P-D-A-G** — Python developer, Data analyst, Automation engineer, Game developer.

COMMON MISTAKE

You do not need all four job titles for the exam. You do need one sentence each if the paper asks "List career opportunities related to this chapter."

اردو میں سمجھیں

اس باب کی مہارتیں پائتھن ڈویلپر، ڈیٹا اینالسٹ، آٹومیشن انجینئر اور گیم ڈویلپر کے کام میں روز استعمال ہوتی ہیں۔ فہرستیں، ڈکشنریز، فنکشنز اور فائلیں چھوٹے اسکول پروگراموں سے لے کر بڑے صنعتی نظام تک ایک جیسی بنیاد ہیں۔

CHECK YOURSELF

1. Which career uses lists, dictionaries and files to interpret business data?
2. Name two careers listed in this chapter.

Answers: 1. Data analyst. 2. Any two of: Python/software developer, data analyst, automation engineer, game developer.

Next lecture: Chapter review (in the chapter lecture PDF)